DIPAN MAZUMDER

AI/ML Engineer | Computer Vision | Full-Stack Developer

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PROFESSIONAL SUMMARY

Machine Learning Engineer specialising in Computer Vision and edge-deployed AI systems. Designed and shipped SARATHI — a real-time driver drowsiness detection system achieving AUC 0.882 at 30+ FPS on Raspberry Pi — and Vrinda, a full-stack agricultural AI platform with 94–96% multi-class accuracy across 15+ crop categories. Experienced with PyTorch, TensorFlow, YOLOv8, BiLSTM, and LangChain across the full ML pipeline from data engineering to production deployment. Finalist at Hack4Bengal 4.0 (7,500+ participants); Winner, SIH Internal Hackathon(400+ participants). Targeting ML/CV engineering and applied AI research roles focused on real-world deployment.

ACHIEVEMENTS

- **Winner** – SIH Internal Hackathon (Smart India Hackathon) among 400+ participants, leading Team 'Dot Slash'.
- **Finalist** – Hack4Bengal 4.0 (7,500+ participants) | Finalist – India Today Technovat Challenge.
- **Global Recognition** – Hack This Fall 2024 AI Project Track (live appreciation, YouTube).

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript (ES6+), HTML5, CSS3

ML / CV: TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, YOLOv8, CNN, BiLSTM, GANs, UNet, VGG19, PhaseNet, Hugging Face, TFLite, RAG

Databases: MySQL, SQLite, MongoDB

Edge / Hardware: Raspberry Pi, TFLite edge optimisation

Frameworks: Flask, Django, FastAPI, Streamlit, REST APIs, Spring Boot

MLOps & Tools: Docker, Git/GitHub, Conda, Google Colab, Kaggle, Postman, VS Code, Roboflow, Netlify

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Pillow, Data Visualisation

Cloud: GCP (Colab), AWS, Vercel, Render

PROJECTS

SARATHI – AI Driver Attention Monitoring System 2024

OpenCV · MediaPipe · BiLSTM · Flask · YOLOv8. github.com/dipanmazumder/sarathi

- Engineered a 4-stage real-time pipeline: YOLOv8n face detection (99.9% detection rate) → MediaPipe FaceMesh (478 landmarks) → 10-feature engineering (EAR, PERCLOS, gaze, head pose) → BiLSTM-Attention temporal classifier, achieving AUC 0.882 on 48-subject subject-disjoint 7-fold cross-validation.

- Implemented per-subject baseline normalisation using first 330 alert frames, enabling personalised drowsiness thresholds that adapt to individual facial geometry — reducing false positives across diverse drivers.
- Deployed as Flask web app with live biometric dashboard and audio-visual alerts running at 30+ FPS on Raspberry Pi; selected as Finalist at Hack4Bengal 4.0 (7,500+ participants).

VRINDA – AI Plant Disease Detection Platform 2024

Node.js · TensorFlow · MongoDB · Express.js · REST API github.com/dipanmazumder/vrinda

- Achieved 94–96% multi-class accuracy across 15+ crop disease categories using TensorFlow CNNs trained on diverse real-world agricultural datasets.
- Built end-to-end full-stack web platform with community forum, fertiliser calculator, weather API integration, and real-time plant health analytics dashboard for farmers.
- Received global recognition at Hack This Fall 2024 AI Project Track with live YouTube appreciation among thousands of international submissions.

Earthquake Phase Detection – Deep Learning System 2024

Python · PyTorch · PhaseNet · ObsPy · SeismoBench github.com/dipanmazumder/earthquake

- Designed real-time seismic event detection using PhaseNet achieving ~97% P/S wave accuracy on the 22 GB ETHZ dataset; achieved 95.5% loss reduction via Gaussian labelling and GPU-optimised training.
- Integrated live seismic-stream ingestion via ObsPy for continuous monitoring, enabling deployment-ready scalability for disaster-response early-warning systems.

WORK EXPERIENCE

Machine Learning Intern Feb 2024 – Mar 2024

Prodigy Infotech *Remote*

- Segmented 200+ retail customers into 5 behavioural cohorts (high-value, conservative, budget-conscious, impulsive, average) using K-Means Clustering with silhouette score of 0.68; identified that top 15% of customers drive 40% of revenue potential, enabling targeted marketing strategy.
- Engineered a house price prediction pipeline on 1,200+ property records using regularised Linear Regression with polynomial and interaction features; achieved R^2 of 0.847 and reduced RMSE by 88% over baseline via GridSearchCV hyperparameter tuning — model serialised with Joblib for production deployment.

EDUCATION

B.Tech in Information Technology Aug 2023 – Jul 2027 (Expected)

Techno International New Town (TINT) *Kolkata, West Bengal, India*

CERTIFICATIONS

- NPTEL: Machine Learning using Python | Certificate 2024
- Google: Data Analytics Capstone | Certificate 2025
- IBM: Generative AI Applications | Certificate 2025